SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Interdisciplinary project in an applied subject (IDP) in Informatics

Implementation of a VR- and LLM-Driven Virtual Patient Interaction System

Yiyang Xie

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Author:	Yiyang Xie
Supervisor:	Prof. Dr. Enkelejda Kasneci
Advisor:	Süleyman Özdel
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Abstract

With the continuous advancement of artificial intelligence (AI) and virtual reality (VR) technologies, digital simulation-based medical education systems have gradually evolved into essential tools for traditional teaching. This transformation holds particular significance in doctor-patient communication training. Clinical practice has long demonstrated that the quality of communication between doctors and patients not only impacts patients' understanding of their conditions but also influences trust, satisfaction, and treatment outcomes.

However, training medical students to communicate with patients more empathetically and to explain complex medical conditions in language patients can understand remains a challenge. Traditional methods, such as standardized patient training and role-playing, while highly realistic, are costly and difficult to organize, further limiting the scalability of teaching.

To enhance the system's scalability and accessibility, this project implements two interaction modes: an immersive first-person perspective mode using VR headsets and controllers, and a non-VR third-person perspective mode controlled via keyboard.

Additionally, this project aims to achieve a balance between immersion, flexibility, and scalability by integrating VR and LLM technologies. Unlike traditional virtual patients that rely solely on scripts or predefined dialogue trees, the patients in this system are LLM-driven and can dynamically generate responses based on user input. Concurrently, the system incorporates an emotional state management mechanism, enabling patients to exhibit emotional shifts in response to conversational content and reflect these variations in their character behaviors.

In terms of implementation, this system uses Unity as its primary development platform, integrating the OpenAI API, the Ready Player Me avatar system, and Mixamo Animation to support virtual patient generation, emotion-driven animation, and real-time conversational interaction.

This paper focuses on the design and implementation of the system architecture, including dialogue state management, emotional modeling, and VR integration. The project results demonstrate that utilizing LLM-driven virtual patients to construct an immersive medical communication training environment is technically feasible. It also provides a foundational system for subsequent expansions, such as automated assessment, expansion of the case database, and user studies.

Keywords: Virtual Reality; Virtual Patient; Large Language Model; Physician-Patient Communication; Medical Education

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1 Introduction

The rapid development of artificial intelligence (AI) and virtual reality (VR) technologies has brought unprecedented changes to education. These technologies show great potential in educational applications.[1] While traditional methods remain effective, engaging students in complex and abstract concepts, such as those in medical education, is challenging. AI systems, especially those powered by large language models (LLMs), offer personalized teaching strategies tailored to individual learning demands. This enables customized learning experiences. Meanwhile, VR technology provides a practical environment where abstract theoretical knowledge becomes visible. This visualization enhances the overall learning experience.[2]

1.1 Background and Motivation

In real-world clinical practice, doctors must possess not only specialized medical knowledge but also solid communication skills. The exchange of information between doctors and patients directly affects the accuracy of the diagnostic process, patients' understanding of treatment plans, and their subsequent therapy adherence. Therefore, the effective cultivation of communication skills among medical students has consistently been a key focus in medical education research.

In traditional medical education, doctor-patient communication training is usually conducted with standardized patients or scenario-based role-playing. In this training, actors, students, or faculty members portray patients. They then engage in simulated dialogues with medical students to help them practice diagnostic interviews and communication skills. This approach can approximate real clinical scenarios to some extent, making it widely used. However, as training scales expand, the model's limitations become clearer. Standardized patient training often requires professional actors, and logistical challenges such as facility arrangements, scheduling, and staff training. These result in high operational costs. Also, differences in actors' availability and performance ability can limit students' access to practice opportunities.

With the advancement of digital technology, an increasing number of studies have begun exploring the use of simulated environments to support medical education. Among these, VR technology has garnered significant attention in recent years.[3] Through VR technology, learners can immerse themselves in simulated clinical scenarios

and perform practical procedures within virtual settings. For instance, in surgical training, clinical procedure practice, and emergency medical drills, VR has been shown to effectively enhance learner engagement and address the limitations of traditional teaching resources.[4]

Meanwhile, advances in LLMs have opened new possibilities for interactive teaching systems. In recent years, LLMs represented by the GPT series have made significant progress in natural language understanding and generation, enabling computers to produce coherent, semantically consistent text. Compared to traditional predefined dialogue systems programmed by developers, LLM-driven conversational agents can deliver more flexible and natural interactive experiences.[5]

Combining VR technology with LLMs has the potential to establish an innovative medical training system. This interactive approach preserves the openness of real clinical communication while overcoming the resource and scale limitations inherent in traditional standardized patient training. Consequently, exploring virtual patient systems that integrate VR and LLMs is emerging as a noteworthy direction in medical education technology research.

1.2 Research Objectives and Structures

This project aims to design and implement a virtual patient interaction system integrating VR technology with LLMs. By leveraging LLM-driven conversational mechanisms, the system enables virtual patients to dynamically generate responses in response to doctors' inputs.

During system design, multiple aspects must be considered at once. For instance, how can a clinical atmosphere be realistically created in a virtual environment? How should virtual patients show emotional changes to make communication feel real? If the system mainly requires VR equipment, its use in standard teaching settings may be limited. Therefore, it is also critical to improve accessibility while keeping immersive experiences.

To address these challenges, the objectives of this study can be summarized as follows:

First, we need to build a virtual patient system that supports natural language dialogue. This will let learners practice communication in ways similar to real clinical interactions. Second, we need to embed mechanisms for emotional states into the virtual patient model. This allows patients to express different emotional responses in response to the doctor's communication style, thereby enhancing authenticity. Third, our system must enable immersive VR interactions so learners can practice in simulated clinical settings. Because VR equipment is limited in real teaching environments, the

project also includes a non-VR mode. This allows users to control characters using the keyboard and to engage in text-based dialogues with virtual patients. Fourth, from a system architecture perspective, the project aims to create a highly scalable framework that can easily accommodate additional case types and scenarios later.

2 Related Work

This chapter reviews the literature on developing an LLMs-powered VR environment for medical education. It focuses on four key areas: virtual reality(VR) in education, large language models in education, traditional medical education, and VR applications in medical education. These areas provide the foundation for understanding current progress and identifying the goals for this project.

2.1 Virtual Reality in Education

VR has emerged as a powerful educational tool, offering immersive and interactive learning experiences that traditional methods cannot replicate. One specific application of VR in education is through "serious games", designed primarily for learning rather than entertainment.

Originally defined as "games for application purposes", serious games are created to impart knowledge, develop skills, provide professional training, and simulate various scenarios [6]. Originating in the 1980s, serious games have been widely applied in multiple fields, including military, healthcare, industry, urban planning, education, training, and scientific exploration. Unlike entertainment games, serious games are characterized by their educational objectives, combining enjoyment and competitiveness with the delivery of practical knowledge and skills [7].

VR is particularly well-suited for serious games because it creates realistic simulations. It can improve understanding and memory of complex information. Makransky et al. found that students who used VR to study complex subjects, such as anatomy, performed better in recall and comprehension than those who used traditional 2D materials [8].

However, integrating VR into education also faces challenges, including high costs, technical issues, and a need for standardized content. For many users, VR can cause motion sickness, as noted by Hoffman et al. [9]. However, technological advancements are gradually overcoming these barriers, which expand the possibilities for VR in education.

2.2 Large Language Models in Education

While VR and AI are distinct concepts, their interaction has become increasingly relevant in recent years. VR provides a simulated experience that immerses users in a virtual world, allowing interaction with virtual objects in realistic environments.

Large language models (LLMs), such as OpenAI's GPT series, have shown significant promise in educational settings due to their ability to process and generate human-like language. LLMs can provide personalized feedback and tutoring across various subjects. LLMs-trained tools are designed to create interactive content, such as Unity6, which is good at generating objects for 3D environments and facilitating scene-based interactions.

Fung et al. highlighted the effectiveness of using LLMs to generate personalized educational content that aligns with learners' needs and preferences, thereby enhancing engagement and improving learning performance [**personalizedlearningpath**]. However, Bashiri et al. noted that LLMs are often used to develop automated grading systems and support educators in administrative tasks, rather than focusing on direct student interaction, feedback, and personalized learning [10].

2.3 Medical Education

Currently, global medical education is undergoing a digital transformation. With the rapid advancement of artificial intelligence (AI), virtual reality (VR)/augmented reality (AR), and various digital simulation technologies, the teaching models and content systems of medical education are being reshaped [11]. The 2024 Medical Education Research Frontiers Analysis indicates that "precision education" and "artificial intelligence" have emerged as core research hotspots [11], while extended reality technologies are increasingly applied in anatomy instruction and surgical training. This wave of technological change not only drives innovation in knowledge transmission but also presents new demands and possibilities for medical humanities and communication skills training.

Among numerous clinical communication skills, effectively delivering "bad news" is regarded as one of the most challenging core abilities. SPIKES [12] is a classic six-step communication model covering: Setting the scene, Perceiving the patient's understanding, Inviting the patient's participation, Sharing medical knowledge, Responding with empathy, and Summarizing and strategizing. Originating in oncology, it has since been widely adopted as the standard framework for doctor-patient communication training.

2.4 VR Applications in Medical Education

VR technology can provide highly immersive learning environments, enabling learners to conduct practical training within virtual spaces [13]. In medical education, VR has been applied to various aspects of medical training, including surgical operations, clinical skills, emergency simulations, and doctor-patient communication.

Research indicates that VR training enhances learner engagement and improves learning outcomes [14]. Compared to traditional role-playing or computer-based 2D simulations, VR technology delivers a more immersive and participatory experience, thereby boosting learning effectiveness [15].

3 Design and Development of the Learning System

This chapter details the development and design of the logic and content of a LLMs-powered medical learning system.

This system comprises multiple interconnected modules responsible for dialogue generation, sentiment modeling, and virtual reality (VR) interaction.

3.1 System Architecture

The system does not centralize all logic within a single script. Initially, we attempted a more direct implementation approach: a central module responsible for receiving input, invoking LLMs, updating state, and triggering animations. However, as functionality expanded, this approach quickly revealed significant maintainability issues. Particularly after supporting both VR and non-VR modes simultaneously, system maintenance became exceptionally challenging.

Therefore, we ultimately adopted a clearer functional layering: user input first enters the dialogue management logic, where the LLM generates patient responses. Simultaneously, the emotion analysis module estimates potential emotional shifts in the patient based on the input and passes the results to the patient state management module. After state updates are completed, these changes further trigger the avatar's distinct behaviors.

3.2 Virtual Patient State Model

The virtual patient system maintains an internal state for each patient object, containing both static and dynamic information.

After the voice guidance, three objects appear:

- **Static data:** Case identifier (ID), personality type, patient symptoms, initial anxiety value

- **Dynamic variables:** Current anxiety value, conversation history

The patient's Anxiety Score represents their current emotional state, ranging from 0 to 1.

3.3 Emotional State Update Mechanism

In real-world doctor-patient interactions, a patient's emotional state is not static. A single remark from the doctor—particularly regarding test results, treatment plans, or risk explanations—can significantly impact the patient's attitude and response.

The system's assessment relies on two emotional signal inputs:

1. Emotional change values returned by the LLM
2. Keyword-based sentiment analysis results

The system combines these two outputs through a weighted average to derive the final emotional state value.

Additionally, to prevent unrealistic emotional fluctuations, the system employs a damping mechanism. When anxiety levels are already high, increasing anxiety becomes more difficult, while decreasing anxiety becomes easier. This behavior approximates real psychological responses.

Listing 3.1: Python Example Code

```
// Principle: The more anxious, the harder to rise further, but easier falls.
    float attenuationFactor;
    if (deltaFromLLM > 0)
    {
        // When anxiety increases: the more anxious, the harder to rise
            attenuationFactor = 1 - current_anxiety;
    }
    else
    {
        // When anxiety decreases: the more anxious, the easier to fall
            attenuationFactor = current_anxiety;
    }

    float adjustedDelta = deltaFromLLM * attenuationFactor;
```

3.4 Dialogue Generation

In a complete human-AI interaction, the user always initiates input as the doctor. In VR mode, this input is processed as voice commands; in non-VR mode, the user inputs text via the keyboard.

Regardless of the method, the system constructs a prompt before invoking the LLM. This prompt does not solely relay the doctor's words to the model. Instead, it integrates multiple types of information, including the patient's current symptoms, personality type, anxiety level, and recent conversation history. The purpose is to ensure the model not only "hears what the doctor has just said" but also understands "the patient's current state" when generating responses.

The model's response is output in JSON format, comprising three key components: the patient's actual reply, anxiety change values, and an indicator of whether the patient comprehends the current explanation. Structured output is preferred over raw natural language text because it enables the system to update the patient's internal state and drive the Avatar's behavior based on this data. If the output were plain text, the system would struggle to reliably link linguistic results with state-update logic.

This approach enables the system to achieve a relatively complete conversational feedback loop.

4 Implementation Process of the Learning System

This chapter details the specific implementation process of the system, including the development environment, key technical tools, methods for constructing virtual characters, LLM-driven dialogue system implementation, and the realization of VR interaction modes.

4.1 Implementation Tools and Development Environment

The development of this system relies on multiple software tools and hardware devices to deliver an immersive, interactive, and personalized virtual patient dialogue experience. The primary development tools are as follows.

4.1.1 Programming Language and Game Engine

This system primarily employs C# as its development language. C# is an object-oriented programming language known for its excellent scalability and stability, widely used in game development and VR application creation.

Since the Unity game engine primarily utilizes C# as its scripting language, it was selected as the system development platform, specifically version Unity 2022.3.12f1 (64-bit).

The decision to avoid newer versions stems partly from the enhanced stability of the Long-Term Support (LTS) release, which better suits academic project development, and partly from plugin compatibility considerations.

The project also integrates Ready Player Me, third-person character control assets, and plugins for language model invocation. To prevent potential compatibility issues during debugging that might arise from overly recent engine versions, we selected a more deterministic, stable, and reliable version over the latest release.

4.1.2 Integrated Development Environment

Visual Studio 2022 is employed as the Integrated Development Environment (IDE) for system development. Within the Unity and C# development workflow, Visual Studio remains one of the most mature and stable solutions available.

Features like syntax highlighting, auto-completion, breakpoint debugging, and error detection are commonplace. In this project, its true value lies in supporting collaborative debugging across multiple scripts. For instance, a failed dialogue may stem from JSON parsing errors, uninitialized patient state objects, or controllers failing to correctly receive parameters. In such scenarios, the IDE's debugging capabilities become crucial.

4.1.3 Large Language Model Interface

To enable natural language interaction with virtual patients, this system integrates GPT-4o's large language model (LLM). This model possesses robust natural language comprehension and generation capabilities, dynamically generating contextually relevant dialogue based on user input.

The system interfaces with the LLM through the OpenAI-Unity plugin. It enables developers to invoke OpenAI APIs in Unity, enabling direct text generation from LLMs in games or VR applications.

At the beginning, we also considered several alternative approaches.

The first approach involved using a locally deployed LLM. However, during medical context testing, the local model demonstrated unstable performance in terminology comprehension, context retention, and adherence to complex prompts, frequently deviating from the intended patient role.

The second approach considered alternative LLMs such as Gemini or Claude. However, based on practical development experience, GPT models proved to be the most mature in content generation while offering the lowest development costs.

4.1.4 Virtual Reality Hardware Devices

For VR hardware selection, this project ultimately adopted the Varjo XR-3 head-mounted display and HTC Vive controllers.

Compared to more common consumer-grade devices like Apple Vision Pro and Meta Quest, the Varjo XR-3 offers superior scalability. For instance, it provides dedicated external cameras for spatial tracking and supports controllers, microphones, and other peripherals from different brands. Thus, the Varjo XR-3 is more suitable for VR simulations in a research context.

Another practical consideration was the experimental environment. The university laboratory was already equipped with Varjo hardware and the corresponding software

ecosystem, including Varjo Base and the SteamVR environment. Continuing to use this platform not only lowered the development threshold but also avoided introducing new compatibility issues associated with hardware replacement.

The HTC Vive controllers primarily handle user input in VR mode. They provide stable hand tracking and button inputs while maintaining strong compatibility with SteamVR.

4.1.5 Virtual Reality Device Management Software

To ensure stable VR hardware operation, the system relies on both Varjo Base and SteamVR software platforms.

Varjo Base primarily handles device-level management, including display configuration, device calibration, and performance tuning. Varjo Base version 4.3 was used during project development, as it was the most stable option for the experimental environment.

On the other hand, SteamVR handles VR runtime management. It serves as the critical bridge enabling Unity to reliably capture headset and controller inputs. Through SteamVR, Unity can read spatial positions, controller movements, and input events, then relay this information to interactive scripts within the scene.

4.2 Virtual Character Construction

Virtual patient characters/Avatars are generated using the Ready Player Me platform. Ready Player Me is an online 3D character creation system that enables rapid creation of virtual character models for use in Unity.

During development, characters are first created on the Ready Player Me platform. They are then imported into the Unity scene by obtaining the GLB model link. The Unity system automatically loads character models by importing the Ready Player Me SDK and integrating it with the Avatar Loader component.

The Unity project imports the Ready Player Me SDK via the following Git repository: `https://github.com/readyplayerme/rpm-unity-sdk-core.git`

The Avatar Loader component is then used to load character models. For example:

- **Female Character:** `https://models.readyplayer.me/6903635663bf032571ed7873.glb`
- **Male character:** `https://models.readyplayer.me/690364a8bc4ae424fc5c9ad6.glb`

Loaded characters can be directly used as Avatars in Unity3D scenes.

4.3 Animation System for Virtual Characters

To enhance the realism of virtual patients, we implemented an animation control system for the characters.

Animation resources are sourced from Mixamo's animation library. Mixamo provides a rich collection of ready-to-use character animations, including talking, angry, and idle animations.

After importing downloaded animations, the first step is to set their Rig type to Humanoid in Unity. This is necessary because the Avatar model utilizes a Humanoid bone structure. After ensuring skeletal compatibility, some animations require editing for specific purposes. Since raw animations often include setup, action, and wrap-up phases, the system may only need a portion of the sequence.

Animation control is managed through the Animator Controller. This system employs controllable trigger logic: when a patient's emotional state reaches a certain threshold, or the system enters a specific speech state, the controller switches to the corresponding animation.

Listing 4.1: Python Example Code

```
// Simple example: Set different animation states based on anxiety levels
        switch (anxietyLevel.ToLower())
        {
            case "mild":
                avatarAnimator.SetFloat("AnxietyLevel", 0.3f);
                break;
            case "moderate":
                avatarAnimator.SetFloat("AnxietyLevel", 0.6f);
                break;
            case "severe":
                avatarAnimator.SetFloat("AnxietyLevel", 1.0f);
                break;
            default:
                avatarAnimator.SetFloat("AnxietyLevel", 0.5f);
                break;
        }
```

In high-anxiety states, the Avatar may exhibit body language such as crossing arms or pointing at the doctor.

Furthermore, this research does not prioritize complex facial expressions for the Avatar. Under current development constraints, stable and consistent body language is more feasible than high-precision facial animation and better aligns with the system's

current objectives.

4.4 Implementation of Dialogue and Emotion Systems

The system's dialogue is powered by ChatGPT-4o. Specifically, the system first receives user input via voice or text. This input is then combined with the patient's current state to form a prompt. The prompt typically includes the patient's symptom description, personality type, current anxiety level, and recent dialogue history.

ChatGPT returns results in JSON format, containing at least three key fields: the patient's response text, anxiety change value, and an indicator of whether the patient understood the doctor's current explanation. This structured output is crucial—if only natural language text were returned, the system could play or display the response, but couldn't integrate the generated content with the emotion update mechanism.

After parsing, the returned results are passed to the patient state object to update the current anxiety value and conversation history. Simultaneously, the emotion update results further drive the animation system.

4.5 Virtual Reality Interaction Mode

In VR mode, users wear headsets to enter a virtual clinic environment and interact with virtual patients from a first-person perspective. Users perform basic interactions via controllers and input speech through microphones. Upon receiving voice input, the system processes the dialogue and feeds the generated responses back to the virtual patient character.

This mode prioritizes authentic communication over complex actions, delivering heightened immersion by making the interaction itself as lifelike as possible.

4.6 Non-Virtual Reality Interaction Mode

In non-VR interaction mode, users control character movement using the W, A, S, D keys on the keyboard, with the system presenting scenes from a third-person perspective. When the character approaches a virtual patient, a dialogue interface is triggered, allowing users to communicate with the patient via keyboard text input.

While this mode sacrifices some immersion, it significantly enhances the system's accessibility. More importantly, the dual-mode design holds another potential value: it lays the groundwork for future comparisons of learning outcomes between immersive 3D-environments and traditional 3D-environments.

4.7 Core Scripts and Operational Logic

To support system operation, this project implements several critical scripts that collectively form the foundational framework for patient state updates, sentiment analysis, language model invocation, and debugging visualization.

- **PatientState:** Maintains all patient state variables and conversation history.
- **AnxietyManager:** Handles anxiety state updates and drives virtual avatar animations.
- **LLMService:** Communicates with the LLM and processes responses.
- **AnxietyAnalyzer:** Performs keyword-based sentiment analysis.
- **AnxietyKeywords:** Configurable keyword dictionary for sentiment analysis.
- **AnxietyDebugger:** Visualization tool for debugging emotional state transitions.

5 Design of Learning Scenarios

This chapter introduces the virtual training scenarios designed within the system, along with dialogue workflow design based on the SPIKES communication protocol. By constructing a virtual environment that replicates authentic clinical settings and integrating LLM-driven virtual patients, the system delivers an immersive patient-doctor communication training experience for learners.

5.1 SPIKES Protocol-Based Communication Learning Framework

The SPIKES protocol is a classic structured framework that physicians use to deliver bad news to patients [12].

Its core components include six key elements. These are: preparing the communication setting (Setting), understanding the patient's knowledge (Perception), inviting the patient to share their desired information level (Invitation), explaining medical information (Knowledge), responding to the patient's emotions (Emotion), and discussing next steps while summarizing the discussion (Strategy & Summary).

This widely accepted framework breaks down the complex communication process into understandable elements. It provides a clear theoretical foundation for doctor-patient interactions [12].

However, this system does not present these six elements as six visible training steps for users, nor does it display them as staged prompts during dialogues. Instead, this research emphasizes a training design featuring implicit staging.

Implicit staging enables learners to engage in a single, natural dialogue with a virtual patient. They do not realize the system organizes training content across the six SPIKES dimensions. The system does not prompt "Now entering the Perception stage" or "Next step: dealing with Emotion." Nor does it force learners to complete steps in a set order. Learners always engage with authentic clinical communication instead of a fragmented, process-oriented task.

This approach mirrors clinical practice and keeps the training immersive. Learners engage with real interactions that often involve complex situations, shifting emotions, and uncertainty.

5.2 Design of Learning Scenarios

To enhance the clinical authenticity and contextual diversity of training, this system features two distinct virtual medical environments:

1. Doctor's Office
2. Operating Room

It is important to emphasize that the distinction between these two scenarios does not correspond to different stages of SPIKES.

Whether in the doctor's office or the operating room, learners consistently engage in a complete doctor-patient communication process that, in theory, encompasses all six elements of SPIKES.

The purpose of scenario design is not to segment training phases, but to influence psychological pressure during communication and the learners' potential emotional state through varying environmental atmospheres, physical layouts, and clinical contexts [16].

5.2.1 Scenario of the Doctor's Office

The doctor's office is a common, private medical setting that fosters trust. This environment visually and spatially resembles a typical consultation room, often even resembling an office.

The interior arrangement includes a desk, swivel chair, sofa, coffee mugs, and potted plants.

Important: This does not mean the scene serves only specific parts of SPIKES. Learners must complete the full communication process. The environment simply supports trust, exploring information, and gentle emotional interaction. The director's office is best seen as a low-pressure setting for communication.

5.2.2 Scenario of the Operating Room

The operating room simulates a more serious, professional clinical environment with heightened psychological pressure.

Unlike the office, this scene clearly shows a medical setting. It features a hospital bed and medical equipment, like scalpels, scissors, syringes, needles, and medication containers.

Compared to the doctor's office, this atmosphere shapes learners' perception of communication tasks. Learners may repeat statements, but the pressure on patients

and doctors can change completely in the operating room. The setting amplifies the seriousness of medical information.

Its purpose is to test and train learners' ability to maintain clear explanations and empathetic expressions within a more challenging medical setting. Furthermore, the dialogue within this scenario is comprehensive and continuous, encompassing all six dimensions of SPIKES.

6 Discussion

This section outlines the contributions and limitations of the system designed in this project to facilitate future development and experimentation.

6.1 Contributions of the System

The core innovation of this study lies in integrating an LLM-driven conversational system with a VR environment to create an immersive, interactive medical communication learning system. Furthermore, the use of SPIKES' implicit learning design avoids over-structuring the communication learning process, preserving its natural flow.

To enhance accessibility, the system design incorporates dual interaction modes: VR and non-VR. This dual-mode architecture enables operation across varying hardware configurations and broadens the system's applicability. At the same time, this design opens new avenues for future research. For example, comparative experiments between VR and non-VR environments could evaluate their respective impacts on learning outcomes. Such studies could further assess the value of immersive technologies in medical education.

6.2 Limitations of the System

Although this study has implemented a comprehensive virtual patient system, the current system still has several limitations.

First, the patient emotion model is relatively basic. This project uses an Anxiety Score as an indicator of the patient's emotional state. The score is dynamically adjusted through LLM feedback and keyword analysis. However, in real clinical interactions, patient emotions are often more complex and can encompass multiple feelings such as anxiety, anger, confusion, or hope simultaneously. Future research may consider incorporating more sophisticated emotion modeling approaches.

Furthermore, this project remains at the system implementation and functional validation stage, without formal user studies. It provides a foundational platform for later work, rather than comprehensive validation of medical learning systems. This shows that medical education can be realized through VR and LLMs. It also provides

the technical foundation for advancing research, including automated evaluation, expanded emotional modeling, and comparisons across different interaction modes.

7 Conclusions and Future Directions

This project designed and implemented a VR- and LLM-driven virtual patient interaction system for doctor-patient communication training. Through this system, learners engage in comprehensive communication within a clinically immersive virtual environment, interacting with a patient whose responses are dynamically generated and whose emotional state evolves throughout the dialogue.

Building on this approach, the system implementation established a comprehensive technical framework, including virtual patient state management, dialogue generation, and emotional analysis. Furthermore, support for both VR and non-VR modes increases scalability and applicability.

On the methodological side, the SPIKES protocol serves as the theoretical foundation for communication training. With implicit phase design, the system maintains natural communication flow, enabling learners to acquire doctor-patient communication skills in full dialogues.

Looking forward, future research can expand system functionality by developing automated communication assessment modules, enlarging the virtual patient case database, and conducting systematic user studies to evaluate practical effectiveness in medical education.

In summary, the proposed system is not intended to replace traditional medical communication training, but rather serves as a supplementary tool. By leveraging VR for scenario creation, LLMs for conversational flexibility, and state-emotion modeling for dialogue continuity and credibility, the system introduces a novel approach. While the full potential of this direction within medical education will be validated through future work, the value demonstrated thus far warrants continued exploration.

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